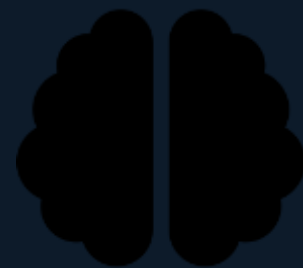
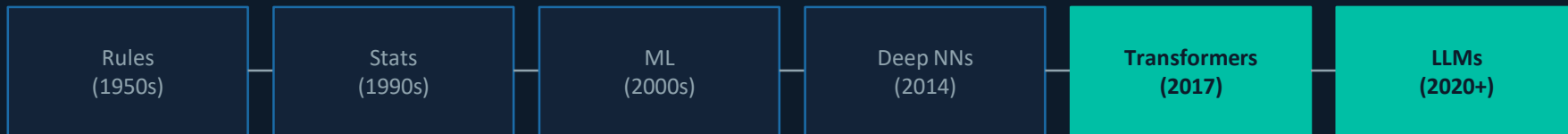


Natural Language Processing



From Classic Algorithms to Large Language Models



Vaswani et al. 2017 · GPT · BERT · ChatGPT · Claude

Syntax & Parsing

Semantics

Transformers

LLMs & ChatGPT

Applications

Why Natural Language Processing?

5.4B

internet users worldwide

328M

tweets per day

720K

hours of video uploaded daily

4.6T

emails sent each day



Machine Translation

Google Translate, DeepL, ChatGPT handle 100+ languages instantly



Search & Retrieval

Google, Bing: rank billions of pages to find the most relevant result



Conversational AI

ChatGPT, Claude, Gemini: chat, summarise, code, answer questions



Sentiment Analysis

Understand opinions in reviews, tweets, customer feedback at scale



Auto Summarisation

Condense long documents, research papers, news articles in seconds



Code Generation

GitHub Copilot, CodeLLaMA write, explain and debug code

What is Natural Language Processing?

Natural Language

- English, French, Bengali, Arabic...
- Evolved organically over centuries
- Ambiguous, context-dependent
- Emotional, creative, nuanced
- Understood through experience

Artificial Language

- Python, Java, C++, SQL...
- Designed by humans, rules are strict
- Unambiguous: one meaning per construct
- No emotion, no metaphor
- Parsed exactly by a compiler

NLP = teaching computers to process, understand, and generate human language

Why Do Computers Struggle with Language?



Lexical Ambiguity

"plant"

✓ She watered the plant. 🌱

✓ He works at the car plant. 🏭

Structural Ambiguity

"I saw the man with a telescope"

✓ I used a telescope to see the man.

✓ I saw a man who had a telescope.

Semantic Ambiguity

"The chicken is ready to eat"

✓ The chicken (food) is ready to be eaten.

✓ The chicken (animal) is hungry.

What HUMANS have:

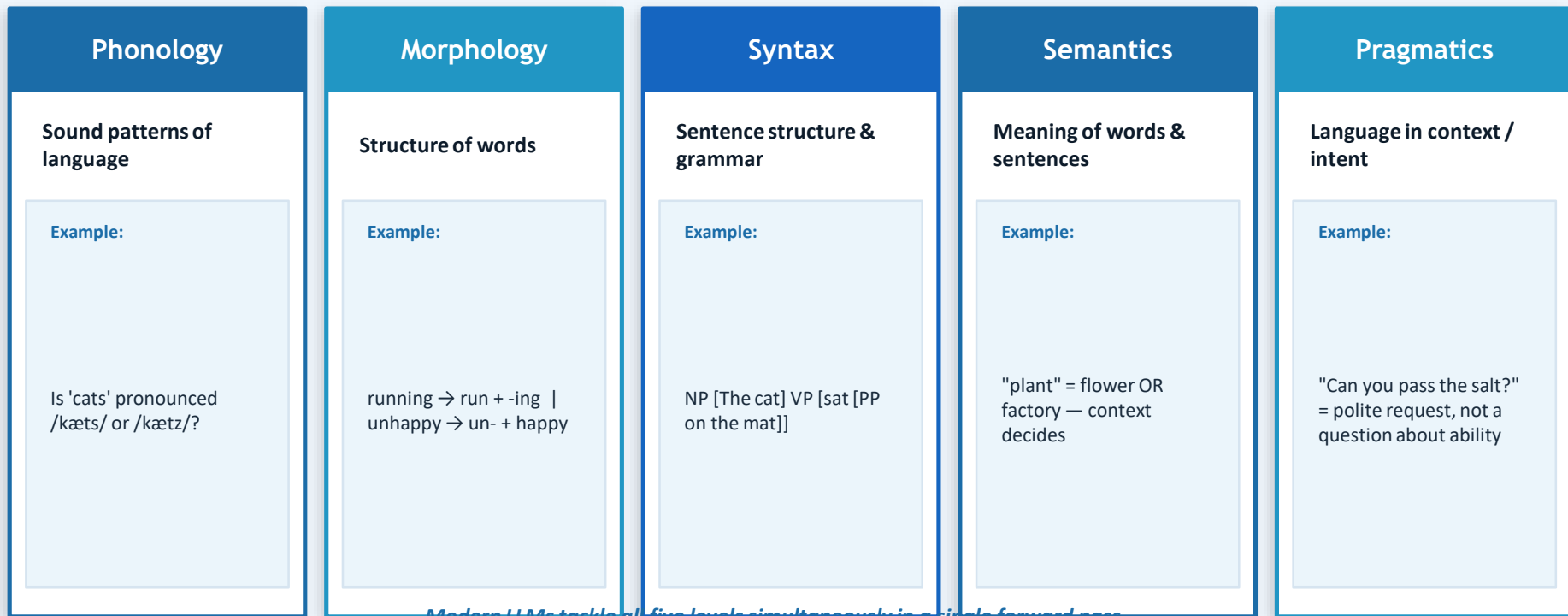
Commonsense knowledge · Real-world experience · Reasoning ·
Context · Emotion

What COMPUTERS had (before LLMs):

Pattern matching · Statistical frequency · Explicit rules · No grounding

Levels of Language — from Sound to Meaning

NLP addresses each level with different algorithms and models



Modern LLMs tackle all five levels simultaneously in a single forward pass

Syntax: POS Tagging, Shallow & Deep Parsing

Part-of-Speech (POS) Tagging



DET=Determiner NN=Noun VBD=Verb(past) RB=Adverb Modern NLP: 98%+ accuracy with neural taggers

Shallow Parsing (Chunking)

Sentence: "The dog chased the bear"

NP [The dog]

VP [chased]

NP [the bear]

→ Identifies basic chunks (subject/predicate)

→ Fast, good enough for many tasks

Full / Deep Parsing

```
S
├─ NP [The dog] ← Subject (Agent)
└─ VP
    ├─ V [chased] ← Predicate
    └─ NP [the bear] ← Object (Patient)
```

→ Full syntactic tree

→ Required for translation, Q&A

Anaphora Resolution & Structural Ambiguity

Two classic hard problems — now largely solved by Transformers

Anaphora Resolution

"The dog entered my room. It scared me."

What does "it" refer to?

Classic approach: gender/number agreement rules

Neural approach: attention over context window

LLM answer: 'It refers to the dog' (correct, instantly)

Prepositional Phrase Attachment

"I saw the man in the park with a telescope."

Two valid parses:

- ① I [used a telescope] to see [the man in the park]
- ② I saw [the man] who was [in the park with a telescope]



How Transformers Solve Both Problems

Self-attention lets every word look at every other word simultaneously. The word "it" computes high attention weight toward "dog" because they share noun/subject properties learned during training on billions of sentences. No explicit rules needed—the model learns coreference resolution from data.

BERT achieves 98%+ on coreference benchmarks | Humans ≈ 99%

Understanding Meaning — Semantics in NLP



Word Sense Disambiguation (WSD) — the word "plant"

Sentence A: "The plant was close to the animal farm."

→ plant = 🌱 vegetation (correct sense)

Sentence B: "The workers went on strike at the plant."

→ plant = 🏭 factory (correct sense)

Dictionary-based (1980s-1990s)

- Look up all senses in WordNet
- Pick the sense that matches nearby words
- Problem: too rigid, misses context

ML Classifiers (2000s-2014)

- 100 human-labelled examples of 'plant'
- Train SVM / Decision Tree
- 60–80% accuracy — limited by labelled data

Word Embeddings + LLMs (2017+)

- word2vec / GloVe / BERT contextual vectors
- Same word → different vector in each context
- GPT-4 achieves near-human accuracy (90%+)

Key insight: context is everything — the same word can mean opposite things. Word embeddings capture this.

Word Embeddings – Giving Words a Mathematical Home

The bridge from symbolic NLP to neural NLP (word2vec 2013, GloVe 2014, BERT 2018)

The Famous Analogy

$$\text{vec}(\text{King}) - \text{vec}(\text{Man}) + \text{vec}(\text{Woman}) \\ \approx \text{vec}(\text{Queen})$$

This means word vectors capture semantic relationships:

$$\text{Paris} - \text{France} + \text{Italy} \approx \text{Rome}$$
$$\text{Doctor} - \text{Man} + \text{Woman} \approx \text{Nurse}$$
$$\text{Swim} + \text{-ing} \approx \text{Swimming}$$

Static Embeddings (word2vec / GloVe)

"plant" always maps to the same vector

- cannot distinguish factory vs flower
- fast, simple, still useful for many tasks

Contextual Embeddings (BERT / GPT)

"plant" in sentence A $\rightarrow$ vector A
"plant" in sentence B $\rightarrow$ vector B
vector A $\neq$ vector B ✓
Context shapes meaning dynamically

Contextual embeddings are produced by Transformers — each word's vector reflects its full sentence context

The Evolution of NLP — Rules to LLMs

1950s–80s	1990s–2000s	2013–2016	2014–2018	2017–2020	2020+
Rule-Based	Statistical NLP	Word Embeddings	RNN / LSTM	Transformers	Large LLMs
• Hand-crafted grammar rules • Expert systems, ELIZA • Brittle: fails on unseen input	• n-gram language models • Hidden Markov Models • Works on data, not rules	• word2vec, GloVe, FastText • Words as dense vectors • Semantic relationships emerge	• Process sequences step-by-step • Handles variable length text • Slow, forgets long context	• Self-attention — parallel! • BERT, GPT-2, RoBERTa • State-of-art on every task	• GPT-3/4, Claude, Gemini • 175B–1T+ parameters • Emergent: reason, code, chat

→ → Each era did NOT replace the previous — it built on top of it → →

The Transformer — Core Architecture

Vaswani et al., 2017 | "Attention is All You Need" | The foundation of all modern LLMs

Self-Attention — Core Idea

"The animal didn't cross the street because IT was tired."

IT refers to → animal (82% attention), not street (3%)

Formula:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) \cdot V$$

Every word attends to every other word — parallel, not sequential

Positional Encoding

- Adds position info (sin/cos waves) — tells model word order without recurrence

Multi-Head Attention

- 8 heads, each learning different relationships (syntax, coreference, semantics...)

Feed-Forward Layer

- $\text{FFN}(x) = \text{ReLU}(xW_1 + b_1)W_2 + b_2$ — processes each word position independently

Add & LayerNorm

- Residual connection + normalisation — allows 6–96 layers without vanishing gradients

Transformer key advantage: parallel computation + long-range dependencies + scales to any size

Transformer Variants — BERT, GPT & T5

Same architecture, different training objectives, different superpowers

BERT

Encoder Only

Google 2018

Training:

Mask 15% of tokens → predict them.
Sees full context (bidirectional).

Best for:

Classification, Q&A, NER, Sentiment

💡 *A student who reads the ENTIRE book before answering any question.*

Params: 110M – 340M

GPT

Decoder Only

OpenAI 2018–2024

Training:

Predict the NEXT word only.
Unidirectional — sees only past.

Best for:

Text generation, Code, Conversation (ChatGPT!)

💡 *A writer composing a story left-to-right.*

Params: 117M → 175B → 1T+

T5

Encoder + Decoder

Google 2019

Training:

Every task = text in, text out.
Unified format for all NLP tasks.

Best for:

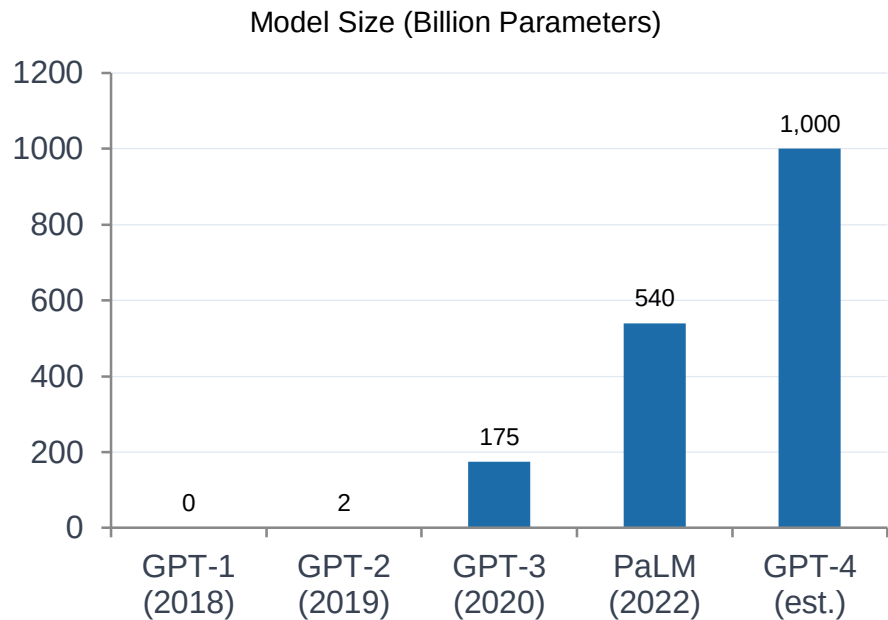
Translation, Summarisation, Multi-task

💡 *A translator: reads source (encoder) → writes target (decoder).*

Params: 60M – 11B

Large Language Models — Scale Changes Everything

GPT-3, GPT-4, Claude, Gemini, LLaMA — and why they feel different



Few-Shot Learning

- Show 3 examples in the prompt → model solves new similar task. No retraining.

Chain-of-Thought

- "Let's think step by step" → model reasons through multi-step maths & logic.

Code Generation

- Describe what you need → working Python/JavaScript/SQL appears.

Cross-lingual

- Trained mostly on English → works well in French, Bengali, Arabic without extra data.

Key insight: emergent abilities appear suddenly at scale thresholds — not present in smaller models at all

Information Extraction & Retrieval – Then and Now

Information Extraction (IE)

"A group of 8-9 people near the entrance on Highway 64."

WHO: 8-9 people (Named Entity: QUANTITY)

WHERE: Highway 64 (Named Entity: LOCATION)

Classic approach: regex patterns + gazetteers

Modern approach: BERT fine-tuned for NER → 93%+ F1 on CoNLL

benchmark

RAG Pipeline (used by Claude, Bing AI, Perplexity):

User query

↓

Retrieve top-k relevant documents (dense retrieval)

↓

Prompt = query + retrieved docs

↓

LLM generates grounded, cited answer

RAG reduces hallucination by grounding LLM answers in retrieved evidence

Information Retrieval (IR)

BM25 (classic)

- Keyword frequency matching — fast, interpretable, still used by Google as a baseline

Dense Retrieval

- Encode query + docs with BERT → compare vector similarity. Finds semantic matches, not just keyword matches.

RAG (2020+)

- Retrieval-Augmented Generation: retrieve relevant docs → feed to LLM → LLM answers with grounded facts.

Machine Translation — A Perfect NLP Success Story



From rule tables to neural systems that rival human translators

Rule-Based MT

- 1950s–80s
- Hand-written grammar rules + bilingual dictionaries.
- Fails on idioms, ambiguity, rare words.

Statistical MT

- 1990s–2010s
- Learn translation probabilities from parallel corpora.
- Google Translate 2006–2016. Good, but misses context.

Neural MT (seq2seq)

- 2016–2017
- LSTM encoder-decoder + attention.
- Huge quality jump. Still struggles with long sentences.

Transformer MT

- 2017+
- Self-attention processes full sentence simultaneously.
- Google Translate switched to Transformer in 2017.

"The cat sat on the mat." → [Transformer] → "Le chat était assis sur le tapis." (French, correct!)

NLP Applications – The Modern Landscape



Language Understanding

- Sentiment analysis (positive/negative/neutral)
- Named entity recognition (WHO WHERE WHAT)
- Question answering & reading comprehension
- Text classification (spam, topic, intent)



Language Generation

- Machine translation (100+ language pairs)
- Automatic summarisation (news, papers)
- Dialogue systems & chatbots (ChatGPT)
- Creative writing & code generation



Multimodal NLP

- Image captioning (describe what you see)
- Visual question answering (GPT-4V)
- Speech-to-text (Whisper) & text-to-speech
- Video understanding with narration



Science & Society

- AlphaFold: protein structure → Nobel 2024
- Clinical NLP: analyse medical records
- Legal AI: contract review, case research
- Education: personalised tutoring (Khan AI)

2017 Transformer paper → 100M ChatGPT users in 60 days (2023) → Nobel Prizes 2024

LLMs: Powerful, But NOT Perfect

Understanding limitations is as important as understanding capabilities



Hallucination

LLMs confidently generate wrong facts. They predict probable text, not true text. No internal fact-checker.

*e.g. "The population of Bangladesh is 2 million."
(Wrong — it's 170M)*



Quadratic Cost $O(n^2)$

Attention computes scores for every token pair. 10,000 tokens = 100M pairs. Long documents are expensive.

e.g. 1000 tokens → 1M pairs | 10,000 tokens → 100M pairs



Pattern Matching ≠ Reasoning

LLMs are sophisticated pattern matchers. They fail on novel multi-step logic not seen in training.

e.g. "If $A > B$ and $B > C$, is $A > C$?" — GPT-3 fails on complex chains.



Bias & Fairness

Models inherit biases from training data. May associate gender with professions, or produce discriminatory output.

e.g. Prompt: 'A doctor walked in. __ picked up the chart.' — biased completion.



Enormous Compute Cost

GPT-3 training cost ~\$4.6M. GPT-4: ~\$100M+. Only a few organisations can train frontier models.

e.g. CO₂ emissions of training one LLM ≈ 5 cars' lifetime emissions.



No Real Understanding

LLMs have no world model, no persistent memory, no genuine comprehension — they are very good at text prediction.

e.g. "Transformers understand language just like humans." — This is FALSE.

Use LLMs critically — always verify important outputs, especially in medical, legal, and educational contexts.

What You Should Know About NLP & LLMs

1

NLP bridges humans and machines

Converting language → numbers (embeddings) is the foundation of everything that follows.

2

Syntax before Semantics

Parsing gives structure; semantics gives meaning. Transformers handle both simultaneously.

3

Context is everything

The same word means different things in different sentences. Contextual embeddings (BERT) capture this.

4

Attention solved the sequence problem

Self-attention processes all words in parallel, capturing long-range dependencies without sequential bottlenecks.

5

Scale brings emergence

Capabilities like reasoning, code, and multi-language appear at scale thresholds — not present in small models.

6

Powerful ≠ Infallible

LLMs hallucinate, inherit biases, and cannot truly reason. Critical evaluation is always necessary.

"Attention is All You Need" (2017) → ChatGPT 100M users in 60 days (2023) → Nobel Prizes (2024)